CHAPTER 17 · DERIVATIONS

MAGNETIC FIELDS

7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CHARGE MOVING ACROSS A UNIFORM ELECTRIC FIELD

SYLLABUS 17.2

RESULT $y = \frac{qEx^2}{2mu^2}$

1. A charge q enters a uniform field E with speed u perpendicular to the field.
2. Along the original direction there is no force: $x = ut$.
3. Along the field the force $qE = ma$ gives a constant acceleration $a = \frac{qE}{m}$: $y = \frac{1}{2}at^2$.
4. Eliminate t: $\mathbf{y = \frac{qEx^2}{2mu^2}}$, a parabolic path (unlike the circular path in a magnetic field, where $F \perp v$ and the speed is constant).

2 FORCE ON A CURRENT-CARRYING CONDUCTOR

SYLLABUS 17.3(B)

RESULT $F = IlB \sin\theta$

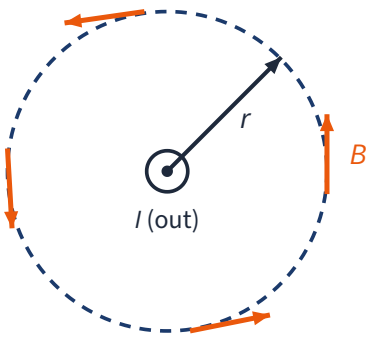
1. Each charge carrier q drifts with velocity v at angle θ to B: force $qvB \sin\theta$.
2. Number of carriers in length l: nAl.
3. Total force $F = (nAl)qvB \sin\theta$.
4. $I = nAqv$, so $\mathbf{F = IlB \sin\theta}$.

3 FIELD OF A LONG STRAIGHT WIRE (AMPÈRE'S LAW)

SYLLABUS 17.4(B)

RESULT $B = \frac{\mu_0 I}{2\pi r}$

1. Ampère's law: $\oint B \cdot dl = \mu_0 I$.
2. Choose a circle of radius r centred on the wire. By symmetry B is tangential and has the same size all round.
3. $\oint B \cdot dl = B(2\pi r)$.
4. $B(2\pi r) = \mu_0 I$, so $\mathbf{B = \frac{\mu_0 I}{2\pi r}}$.



4 FORCE BETWEEN TWO PARALLEL CONDUCTORS

SYLLABUS 17.5(A)

RESULT $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$

1. Field of wire 1 at wire 2 (distance d): $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.
2. Force on length l of wire 2: $F = B_1 I_2 l$.
3. $\mathbf{F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}}$; an equal and opposite force acts on wire 1.
4. Currents in the same direction attract; opposite directions repel.

5 E/M IN THOMSON'S EXPERIMENT

SYLLABUS 17.6(C)

RESULT $\dfrac{e}{m} = \dfrac{E}{rB^2}$

1. Crossed E and B fields are adjusted so the beam is undeflected: $eE = evB$, so $v = \dfrac{E}{B}$.
2. With E switched off, the electrons move in a circle of radius r in B: $evB = \dfrac{mv^2}{r}$.
3. $\dfrac{e}{m} = \dfrac{v}{rB}$.
4. Substitute $v = \dfrac{E}{B}$: $\mathbf{\dfrac{e}{m} = \dfrac{E}{rB^2}}$.

Alternative speed selection by energy: $eV = \tfrac{1}{2}mv^2$, so $v = \sqrt{\dfrac{2eV}{m}}$ and $\dfrac{e}{m} = 2V/(B^2r^2)$. State which condition you use.

6 BAINBRIDGE MASS SPECTROMETER

SYLLABUS 17.6

RESULT $m = \dfrac{qBB'r}{E}$

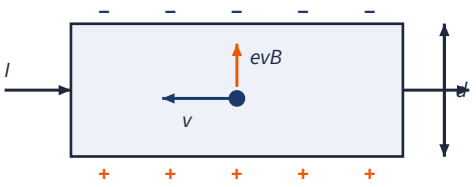
1. **Speed selector**: crossed E and B fields; only ions with $qE = qvB$ pass undeflected, so $v = \dfrac{E}{B}$.
2. **Deflection chamber**: a second field B′ alone; the magnetic force provides the centripetal force: $qvB' = \dfrac{mv^2}{r}$.
3. $r = \dfrac{mv}{qB'} = \dfrac{mE}{qBB'}$, so ions of different masses (isotopes) follow semicircles of different radii.
4. $\mathbf{m = \dfrac{qBB'r}{E}}$, or specific charge $\dfrac{q}{m} = \dfrac{E}{BB'r}$; the separation of two isotopes on the plate is $2\Delta r$.

7 HALL VOLTAGE

SYLLABUS 17.7(B)

RESULT $V_H = \dfrac{BI}{net}$

1. Carriers (charge e, number density n) drift with velocity v through a slab of width d and thickness t, with B perpendicular.
2. The magnetic force evB pushes carriers to one face; charge builds up and sets up a Hall field $E_H = \dfrac{V_H}{d}$.
3. Equilibrium: $\dfrac{eV_H}{d} = evB$, so $V_H = Bvd$.
4. $I = neAv = ne(dt)v$, so $v = I/(nedt)$.
5. $\mathbf{V_H = \dfrac{BI}{net}}$.



B into page; V_H across *d*; thickness *t*

Condition: equilibrium ($F_{electric} = F_{magnetic}$) while the current continues to flow. $V_H = Bvd$ needs the width d; in $V_H = \dfrac{BI}{nqt}$ the Hall voltage depends on the thickness t, not on the width.